



Distribution of polycyclic aromatic hydrocarbons (PAHs) from legacy spills at an Alaskan Arctic site underlain by permafrost



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ABSTRACT

Widespread polycyclic aromatic hydrocarbon (PAH) contamination resulting from petroleum spills at military and industrial sites along the Arctic Ocean Coast is a major issue in the Alaskan Arctic. Polycyclic aromatic hydrocarbons (PAHs) consist of one or more fused aromatic rings and comprise a major portion of petroleum products like fuels. PAHs are commonly found in a variety of geomedias and are considered environmentally persistent, toxic, and carcinogenic. In the present study, the subsurface distribution of PAHs was investigated at a former U.S. Department of the Navy site in northern Alaska using laser-induced fluorescence coupled with ultraviolet optical screening tool (LIF-UVOST). The former Naval Arctic Research Laboratory (NARL) study site has historical presence just outside the city of Utqiagvik, Alaska (formerly Barrow) and has a long history of petroleum spills since the early 1950s primarily at the Airstrip and Powerhouse sites. To determine the extent of PAH accumulation in the subsurface a total of 143 vertical soil profiles were characterized using LIF-UVOST during September 2015 (87 at the Airstrip site and 56 at the Powerhouse site). We also compared the intensities of different wavelengths (350 versus 500 nm) to estimate spill source and we found a substantial accumulation of PAHs in the surface and subsurface at both sites and vertical distribution of PAHs was primarily heterogeneous, varying as a function of probing locations and soil depth. Both sites were found to have no correlation between depth of the maximum fluorescence signal and depth of the probing refusal depth (top of permafrost). Using LIF-UVOST for the direct chemical sensing of PAHs proved to be a useful tool and revealed unique signature PAH fluorescence responses when comparing 350 and 500 nm intensities between the Airstrip and Powerhouse sites, likely attributable to the original source of the spill. Results from this project can be directly used for other potential studies, including planning remediation strategies and investigating local bodies of water for PAH transport.

1. Introduction

The occurrence of petroleum, oils, and lubricants (POLs) in soil and groundwater systems as a result of spills is a major human health concern and an environmental risk to land sustainability. Petroleum-based hydrocarbons are used for a variety of applications including fuel, fuel additives, solvents, thinners, and the production of petrochemicals. As a result, widespread usage of petroleum and petroleum products has caused fast-growing incidents of spills. Introduction routes for petroleum in the environment are often due to point-source spills, pipeline breaches, and storage tank leakages (Balseiro-Romero et al., 2016; Christophersen et al., 2005; Foong and Rahman, 2017; Stout, 2003; Wang et al., 2006). However, dissolution and groundwater infiltration are considered dominant pathways that pose the greatest risk

to the ecosystem and human health (Johnston and Trefry, 2009). In the Arctic, the presence of restricted flow regimes from permafrost, a short annual thaw season, and the dominance of sub-zero temperatures presents added barriers to evaluating distribution of petroleum-based hydrocarbons. As such, investigating the fate and transport of petroleum-based hydrocarbons in the surface and subsurface soil is a major research area in cold regions, with particular importance for planning remediation strategies.

Polycyclic aromatic hydrocarbons (PAHs) comprise a major portion of petroleum products which are in general a complex mixture of hydrocarbons, organic compounds, and also contain some heavy metals (Speight, 2014). The toxicity of PAHs has been well characterized during the last few decades and numerous ecological and environmental procedures have been established (ATSDR, 1996; Nisbet and

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Lagoy, 1992). PAHs have been shown to be carcinogenic, teratogenic, and mutagenic as a result of the presence of two or more benzene rings fused together (Dong et al., 2015). In general, PAHs have a low aqueous solubility, which tends to decrease with increasing molecular weight and addition of rings (Abdel-Shafy and Mansour, 2016). In the environment, PAHs persist and can partition into mobile dissolved and vapor phases, which can cause widespread migration on a relatively short timescale (Johnston and Trefry, 2009). Once PAHs are released into the environment as a result of a land-based spill the fate and transport is largely controlled by soil properties, speciation of PAH, presence or absence of soil organic material, and hydrological flow paths (Abdel-Shafy and Mansour, 2016; Cachada et al., 2012; Riccardi et al., 2013).

Previous studies on PAHs in the environment often focus on fluorescence detection in oceanic surface waters due to the prevalence of petroleum-based products spilling from oil tankers (Driskill et al., 2018; Sinaei and Mashinchian, 2014). For land-based studies, many have solely focused on quantifying the concentrations of PAHs in soil and groundwater, often using ‘grab’ samples (Cachada et al., 2012; Riccardi et al., 2013; Zhang et al., 2006). While quantifying concentrations in soil and groundwater can yield a general idea of the degree to which a site is contaminated with PAHs, it says little about the in-situ vertical distribution of PAHs and how the concentrations relate across a given site. Understanding the extent to which PAHs have migrated is important for detailed site-wide evaluations, characterizing the PAHs or other petroleum-derived hydrocarbons present, and planning bioremediation or inorganic remediation strategies. In this study, we analyzed the in-situ vertical distribution of PAHs in soils and compare the results across two different Arctic coastal sites.

The major hydrocarbons that comprise petroleum are made up of aromatic rings that can absorb energy at wavelengths in the 350–500 nm range and produce characteristic fluorescence patterns, which can be useful in identifying unique PAHs (Pena et al., 2015; Rivera-Figueroa et al., 2004; Suthersan et al., 2017). Fluorescence spectroscopy represents an attractive method for the in-situ detection of PAHs in the subsurface because it has been shown to be rapid and reliable (Driskill et al., 2018; Suthersan et al., 2017; Wu et al., 2014) and has been successful in detecting fluorescence species in water, organic matter, and soil (Martins et al., 2011; Muller et al., 2011; Uebel et al., 1996). Standardized methods of the detection of signature PAH fingerprints found in common petroleum-based products have shown distinct emissions that can be characterized based on the intensities of fluorescence as a function of wavelength (350, 400, 450, and 500 nm). There are several commercially available fluorescence screening tools that rely on detection of the fluorescence of non-aqueous phase liquids (NAPLs) when they are exposed to induced fluorescence. One example is the laser-induced fluorescence Ultra Violet Optical Screening Tool (LIF-UVOST) developed by Dakota Technologies (St Germaine, 2011) utilizes LIF to detect the presence of polycyclic aromatic hydrocarbons (PAHs) using these four wavelengths. Ultraviolet light is transmitted through an optical wire that exits through a sapphire window on the bottom of the probe tip. The laser light excites the PAH fraction into releasing energy and fluorescence. The intensity of the fluorescence is used as an indicator of the relative concentration of contaminants. LIF results are recorded in real time with depth and presented as a graph of fluorescence intensity versus depth. UVOST detects petroleum, oils and lubricants (POL), gasoline, diesel fuel, biodiesel, motor oil, cutting fluids, hydraulic fluid, and jet fuel. More recently a new system to extend the capability to detect dense non-aqueous phase liquid (DNAPL), the DyeLIF, has been developed (Einarson, 2016; Einarson et al., 2018). LIF systems have been performed for high-resolution mapping of petroleum hydrocarbon containing PAHs (e.g. Bujewski and Rutherford (1997) and Aldstadt et al. (2002)). Wu et al. (2014) successfully demonstrated that LIF can be used as a semi-quantitative method to source the type of petroleum spill. PAH can also be quantified with a number of analytical methods including gas chromatography- mass

spectrometry (GC–MS) or GC coupled to a flame ionization detector (GC-FID) and high performance liquid chromatography (HPLC) with fluorescence detection. These techniques primarily employ sample preservation techniques and extraction methods and are typically analyzed in laboratory setting and not in-situ like LIF-UVOST. Additionally, the U.S. Environmental Protection Agency (USEPA) has previously named UVOST as a reliable method for expedited site assessments for sites where petroleum is stored (USEPA, 2016).

In the current study, the subsurface distribution of PAHs was investigated at a well-characterized site located in northern Alaska using LIF-UVOST mounted to a push drill rig. The study site investigated was the former Naval Arctic Research Laboratory (NARL), which has a long history of petroleum spills since the early 1950s. Characterizing the distribution of PAHs in the soil at the former NARL site is particularly important as it is located adjacent to a legacy, and secondary local drinking water source (Imikpuk Lake) for the city of Utqiagvik. Overall, a total of 143 vertical soil profiles were investigated using LIF-UVOST during September 2015. Additionally, the probing refusal depths were determined by push drill rig refusal detection to determine if there was any association with probing refusal depth and PAH accumulation as this has implications for frozen barriers and contaminant fate and transport. The present case study focuses on obtaining baseline results to investigate the horizontal and vertical extent of petroleum fuel and oil contamination at the Airstrip and Powerhouse sites of the former NARL complex using LIF-UVOST capabilities.

2. Materials and methods

2.1. Field study

The Naval Arctic Research Laboratory (NARL) was established at Point Barrow, Alaska, outside the city of Utqiagvik in 1947. The city of Utqiagvik is located approximately 6 km southeast from the Airstrip runway of the former NARL complex (Fig. 1). The former NARL site and Utqiagvik are located along the northern edge of the Teshekpuk Lake section of the Arctic Coastal Plain physiographic province of Alaska. The NARL facility officially closed in 1980 and was decommissioned in 1981. Over the life of NARL, over 1 million liters of fuel was documented as spilled. An overview of historical monitoring and remediation efforts are documented by Bjella et al. (2018). The Ukpeagvik Inupiat Corporation (UIC) overtook the operation of the laboratory site in 1984 and the majority of NARL was transferred to UIC as part of a land exchange agreement in 1986 and used as a facility support center for local activities (Crowser, 2000a; Crowser, 2000b; McCarthy et al., 2004; Reed and Ronhovde, 1971; Shelesnyak, 1948). Our two study locations are located on property of the former NARL complex and include the Airstrip Site and the Powerhouse Site (Fig. 1).

The former NARL complex lies in within an area of continuous permafrost and permafrost depths have been known to exceed 250 m in some cases (Osterkamp and Payne, 1981). Typical land features in the area include shallow marshes, thaw lakes, braided meandering rivers, beaded streams, scattered sand dunes, ice-wedge polygons, and pingos (Zhan et al., 2014). The primary geologic composition is marine and non-marine deposits, consisting of unconsolidated silts and sands with some clays and gravels also with shale, mudstone, and sandstone (Black, 1964; Payne et al., 1951; URS, 1995). Organic and clay or loam deposits of tundra meadows overlays the gravel deposits at some locations. Sand and gravel were placed over the tundra as NARL was expanded. These sand/gravel fill layers are on average about 0.5 to 1.2 m but could be up to 3 m thick (URS, 1995). Precipitation in the summer typically occurs as numerous light showers or heavy mists. In the winter, the majority of precipitation is snowfall (Reed and Ronhovde, 1971). Mean monthly temperatures are estimated to range from -46°C to 7°C with an annual average of -12°C (Shulski and Wendler, 2007). The primary vegetation in this area is tundra vegetation, consisting of sphagnum mosses, grasses, sedges, and lichens and

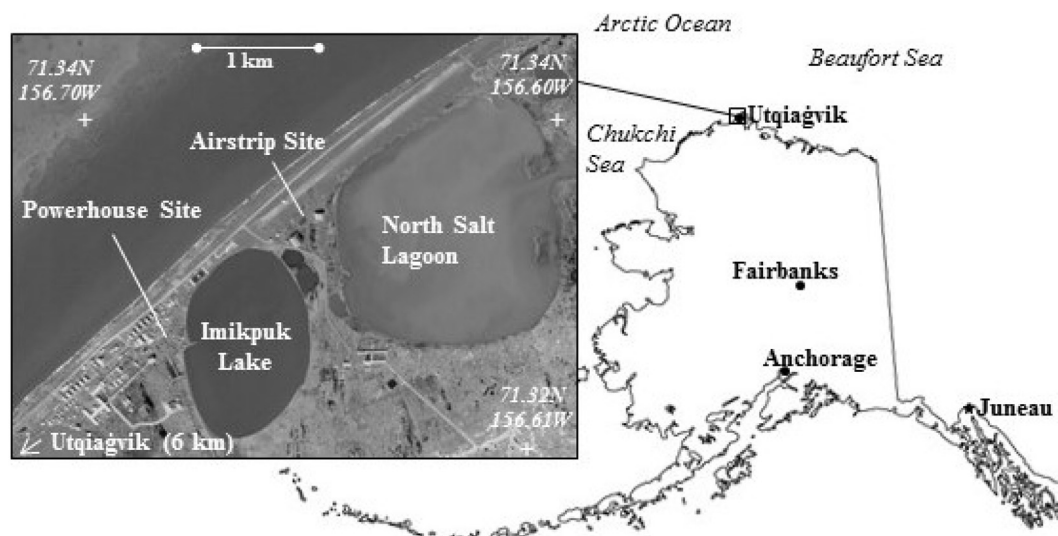


Fig. 1. Map of the study area with sampling sites (Airstrip and Powerhouse site) identified. Accessed using Google Earth V 7.3.1.4505 (January 17, 2018) on January 30, 2018, “Utqiagvik, Alaska”. 71°17′25.95″N, 156°47′19.01″W.

little to no trees besides sparse distributions of small willows and small alders (Reed and Ronhovde, 1971). However, due to the installation of buildings and a runway at the complex there is little vegetation located directly on-site. The thaw season (early/mid-June to late September) in Arctic Alaska yields high, relative runoff during spring freshet and shallow, transient groundwater tables (McCarthy and Solin, 1995; McCarthy et al., 1994). Annually, the majority of the area experiences freezing conditions and stream flow is minimal or does not occur at all. Therefore hydrological processes are primarily controlled by accumulation of snow, sublimation from the snowpack, and redistribution of soil moisture (Braddock and McCarthy, 1996; Douglas et al., 2017; McCarthy et al., 1994).

2.1.1. Airstrip site

The Airstrip site is located about 6 km north of Utqiagvik and approximately 250 m inland from the shoreline of the Arctic Ocean in Alaska (Fig. 1). The Airstrip site is owned by the U.S. Navy and covers about 500,000 m² and includes a 1520 m runway, a large hangar (Hangar 136), an apron area, and other buildings. The runway and apron areas are surfaced with jointed steel plate matting (Marston Matting) and the apron area has two layers of this matting.

The primary contaminants of concern have been documented at the site as a result of known fuel spills over the active history of the former NARL site. Spills have included gasoline range organics (GRO), diesel range organics (DRO) and volatile organic compounds (VOCs). During historical remediation and cleanup efforts PAHs were found in some of the soil and liquid samples, further outlined in Bjella et al. (2018). Approximately 1,386,000 l of fuel spills are documented during the Airstrip operational period. Gasoline and jet petroleum No. 5 (JP-5) fuel spills occurred on the site between 1976 and 1986. In 1976, about 183,000 l of gasoline was spilled due to an underground fuel line failure. An additional 1,050,000 l of gasoline was spilled in 1978 where about 530,000 l of usable gasoline was recovered. In 1986, approximately 61,000 l of JP-5 was released and about 4000 l were recovered the following year.

2.1.2. Powerhouse site

The Powerhouse site lies approximately 300 m inland from the shoreline of the Arctic Ocean, northeast of the city of Utqiagvik in Alaska (Fig. 1). The Powerhouse site includes Building 342 (former Powerhouse), Building 442 (inactive water pump house), and the Ukpægvik Iñupiat Corporation (UIC) Construction equipment lay-down area. At least 95,000 l of JP-5 was released at this site between 1952

and 1988. The primary contaminants that have been detected at the Powerhouse site are GROs, DROs, VOCs, and residual range organics (RROs) (Linder, 1998).

2.2. Data collection

Field work and UVOST probing surveys were conducted August 16 through September 01, 2015. In total, 143 UVOST probes were conducted including 87 UVOST probes performed at the Airstrip site and 56 UVOST probes at the Powerhouse site. A UVOST probe (Dakota Technologies, Fargo, ND) was attached to a direct push drill rig (Geoprobe Model 6610DT, Salina, Kansas) and advanced through the soil column at various locations throughout the Airstrip and Powerhouse sites at a rate of approximately 22 mm/s. The UVOST probe is comprised of fiber optic cables that emit pulsed ultraviolet laser light (Excimer laser source, Dakota Technologies, Fargo, ND) at four different wavelengths (350, 400, 450, and 500 nm) that is reflected by a mirror through a fitted, transparent, sapphire window into the adjacent soil column. The mirror is metallic, parabolic, and optically-pure and constructed by Dakota Technologies (Fargo, ND). The UVOST probe is equipped with a fluorescence detector (Dakota Technologies, Fargo, ND) at the surface that is connected to a separate fiber optic line and data is collected in real-time. The UVOST probe was calibrated by applying a reference fluid standard to the window and recording response prior to each fluorescence log. The reference fluid standard is comprised of a stable, fluorescent NAPL that is a proprietary mixture of standardized oil products (Dakota Technologies, Fargo, ND). The bottom of the seasonally thawed “active layer” (top of the permafrost depth) was also noted by refusal during UVOST drilling operations. However, the exact depth to permafrost proved to be difficult to obtain quantitatively from the probing due to the prevalence of compacted sand at both sites and the presence of Marston Matting on the Airstrip Runway. It is estimated that the probing refusal depths reflect the top of the permafrost to within approximately ± 20 cm based on repeated measurements.

Survey points for the study were chosen to accurately assess the extent of petroleum spills determined from legacy information at the Airstrip and Powerhouse sites, with a particular focus on probing locations near bodies of water. Each UVOST point was surveyed using a Trimble R8 survey-grade global positioning system (GPS) with reported positional accuracies < 2 cm. The coordinate system used to collect, process, and display survey data in this study is WGS 84 UTM Zone 4 N. The vertical control datum NAVD88 is used to establish ground

elevations.

2.3. Data analysis

Fluorescence data was analyzed using Optical Screening Tool (OST) software (Dakota Technologies, Fargo, ND). Fluorescence intensity (pVolt/s) at each wavelength of energy emitted (350, 400, 450, and 500 nm) was normalized by dividing the total fluorescence intensity by the fluorescence intensity of the reference standard at each wavelength of energy to achieve relative percentages of signal intensity. The fluorescence contribution of each individual wavelength of photons emitted was obtained by summing the area under the fluorescence signal curve. This process was only done for the depths corresponding to the highest fluorescence signal detected in each drilling. Background noise intensity was determined relative to the reference standard and collected for each fluorescence signal. Plume contour maps were constructed for each of the sites using the UVOST probing data by interpolating the scattered set of fluorescence data by location with inverse distance weighting (IDW) of value averages with at least 10 nearest probing locations within ArcGIS (ESRI, Redlands, CA).

3. Results and discussion

3.1. Distribution of PAHs

The spatial distribution of the PAH signals detected at the former NARL site revealed multiple locations exhibiting significant fluorescence intensity as a result of UVOST probing. The plume contour maps for Airstrip and Powerhouse sites (Figs. 2 and 3) show the locations of the maximum fluorescence intensity (%) detected by UVOST probing and were constructed by inverse distance weighting (IDW) interpolation, including at least 10 nearest probing locations. The Airstrip site shows significant PAH accumulation directly under the Runway and the Runway Apron (probing sample A9, Fig. 2). Probing A9 and A45 account for the probing locations with the highest fluorescence intensity detected at the Airstrip site (71% and 30%, respectively). Within the

contour map, probing location A9 does not necessarily reflect that all surrounding area is > 12.5% total fluorescence response, but that the high, relative fluorescence signal (71%) encountered at A9 impacts the IDW interpolation of the surrounding area. From the Airstrip Runway to Imikpuk Lake, the plume contours decrease in fluorescence intensity of PAHs with other locations exhibiting elevated, relative intensities of PAHs located closer to Imikpuk Lake (A26, A27, A34, A51–52, A55, and A59). There are two probing locations that are located east (A70; Fig. 2) and southeast (A80; Fig. 2) of the Airstrip Runway where PAHs intensity corresponded to higher than background levels and both are located near Dewline Road.

The Powerhouse site contained a larger amount of probing locations with high, relative PAH fluorescence intensities (Fig. 3) than the Airstrip site. There were five probing locations (P26–28; P46 and P51) that are directly west of Imikpuk Lake that contained maximum fluorescence intensities (%RE) > 12.5% and seven probing locations (P7, P12–13, P17–18, and P40–41) distal to Imikpuk Lake closer to Buildings 140, 342, and 442 that showed a maximum fluorescence intensity (%RE) of > 12.5%. The probing locations in between the two areas with %RE values above 12.5% show plume contours primarily between 2.5 and 12.4% fluorescence intensity. Fluorescence intensities from substances termed as ‘false positives’ can occasionally be an analytical issue using LIF-UVOST detection methods for PAH distributions in soils, primarily arising from interferences with seashells, wood, peat, and calcareous sands. However, interference response from a false-positive substances generally yields both a low intensity fluorescence signal and the integrated area under the curve for each wavelength investigated (350, 400, 450, and 500 nm) tends to be small and non-consistent (Suthersan et al., 2017). Therefore, any incorporation of fluorescence signal as a result of a non-PAH containing substance is considered negligible for this study based on the high, relative fluorescence responses of the majority of the probing locations.

The vertical thicknesses in relation to permafrost refusal depths and soil depth locations of the maximum fluorescence intensities were investigated at the two study sites (Figs. 4 and 5). The majority of PAH intensities were encountered at the Airstrip site between the surface

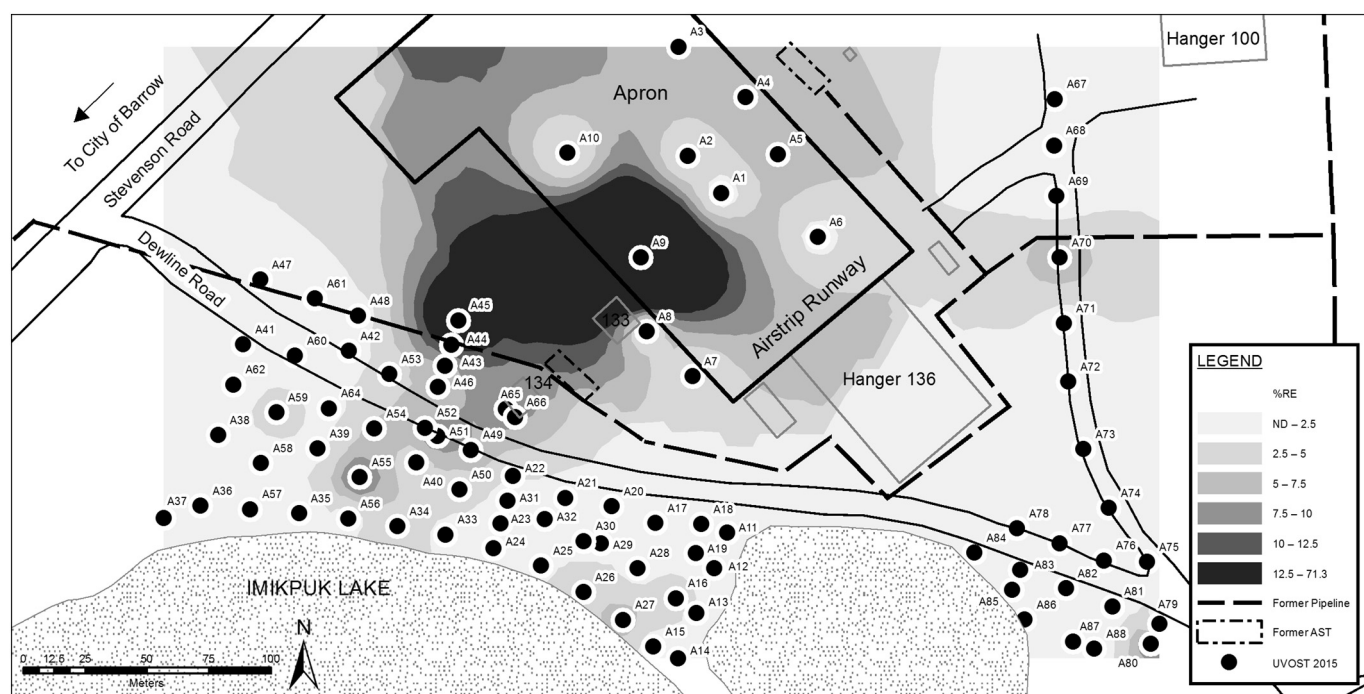


Fig. 2. Plume contours for the Airstrip site showing maximum %RE (response) values between ND (not detected) and 71.31% RE. Plume contours were made by assigning the %RE max at each probe location and contouring value until the next adjacent probe location maximum %RE value. Probing locations, former pipeline, and former above-ground fuel storage tanks (AST), and existing structures shown.

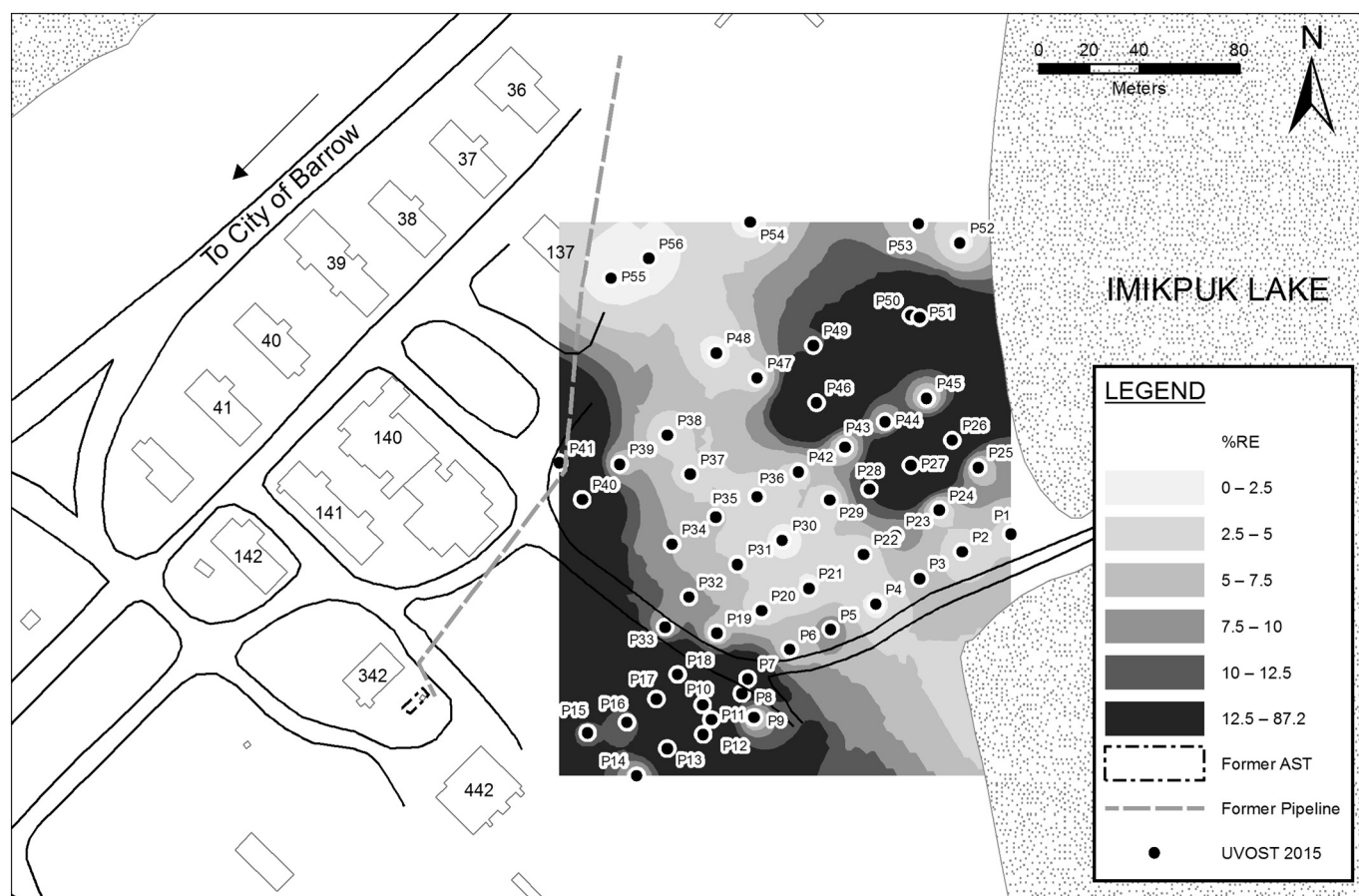


Fig. 3. Plume contours for the Powerhouse site showing maximum %RE (response) values between ND (not detected) and 87.2% RE. Plume contours were made by assigning the %RE max at each probe location and contouring value until the next adjacent probe location maximum %RE value. Probing locations, former pipeline and above-ground fuel storage tanks (AST), and existing structures shown. Note: P50 exhibited a %RE max of 3.9% instead of > 12.5%, which is shown on the map. The IDW-weighted averages could not account for large differences in significantly close proximities.

and 2.0 m depth (Fig. 4), with the exception of probing location A9 located directly on the Airstrip Runway Apron (Fig. 2). Probing location A9 exhibited PAH intensities deeper in the subsurface from approximately 2.2 to 3.2 m. The probing location with the largest vertical PAH thickness was A55, which is located directly between Dewline Road and Imikpuik Lake (Fig. 2). The average maximum response depth across the Airstrip site occurred at 0.9 m depth with a standard deviation of 0.5 m. In general, the extent of the vertical thicknesses of the PAH plume at the Airstrip site were within 1.0 to 2.0 m soil depth.

The majority of PAH intensities were encountered at the Powerhouse site between the surface and 2.0 m depth (Fig. 5), with the exception of probing location P13, which is located between Imikpuik Lake and Building 442 (Fig. 3). The probing location with the largest vertical PAH thickness was also P13. The average maximum response depth across the Powerhouse site occurred at 0.6 m depth with a standard deviation of 0.4 m. In general, the extent of the vertical thicknesses of the PAH plume at the Powerhouse site were within 0.5 to 2.0 m soil depth. In comparison to the Powerhouse site that contained probing refusal depths generally at or above 2.0 m, the Airstrip site generally contained sampling locations with deeper, relative probing refusal depths. In particular, sampling locations A18–19 exhibited probing refusal depths deeper than 4.0 m and sampling locations A9, A11, A15, and A17 exhibited probing refusal depths deeper than 3.0 m (Fig. 4). The maximum probing refusal depth at the Powerhouse site was slightly above 3.0 m depth (Fig. 5).

The fluorescence data collected at the Airstrip and Powerhouse sites at the former NARL complex clearly show evidence of large-scale petroleum spills and support the known historical locations, including the

Airstrip Runway (Fig. 2) and the location of the Former aboveground storage tank (AST) (Fig. 3). Otherwise, the vertical PAH spread for the Airstrip and Powerhouse sites show that petroleum-derived contamination is largely heterogeneous with respect to vertical and spatial distribution. Sampling locations in close proximity to each other (~1–5 m) showed often significantly different PAH distributions. For example, the Airstrip sampling locations A27, A55, A59, and A88 contained high, relative fluorescence signals compared to the surrounding probing locations (Fig. 2). The majority of the PAHs located at A27 are within the first 1.0 m of the soil profile, whereas the PAHs are distributed deeper at probing locations A55 (Fig. 4).

In a parallel study performed by Barbato et al. (2017), PAH accumulation was found to be heterogeneous as a function of study site and soil depth. Barbato et al. (2017) investigated soil microbial activity and found that concentrations of aromatic hydrocarbon fractions varied greatly (Barbato et al., 2017). During this study grab samples were collected from the Airstrip and Powerhouse at a total of 16 sites and analyzed using a gas chromatograph equipped with a flame ionization detector (GC-FID). Maximum aromatic hydrocarbons (carbon chain length: C21–C34) at the Airstrip site ranged from below the detection limit of the analytical method (further outlined in Barbato et al. (2017)) to almost 580 mg/kg-dry and similarly at the Powerhouse site the concentration ranged from non-detect to about 1150 mg/kg-dry. Results from the Barbato et al. (2017) study show similar quantitative aromatic hydrocarbon response as measured UVOST intensities. However, no direct comparison between the two analytical techniques (UVOST and GC-FID) was made except to confirm presence or lack of PAH presence. No aromatic hydrocarbons were detected using GC-FID

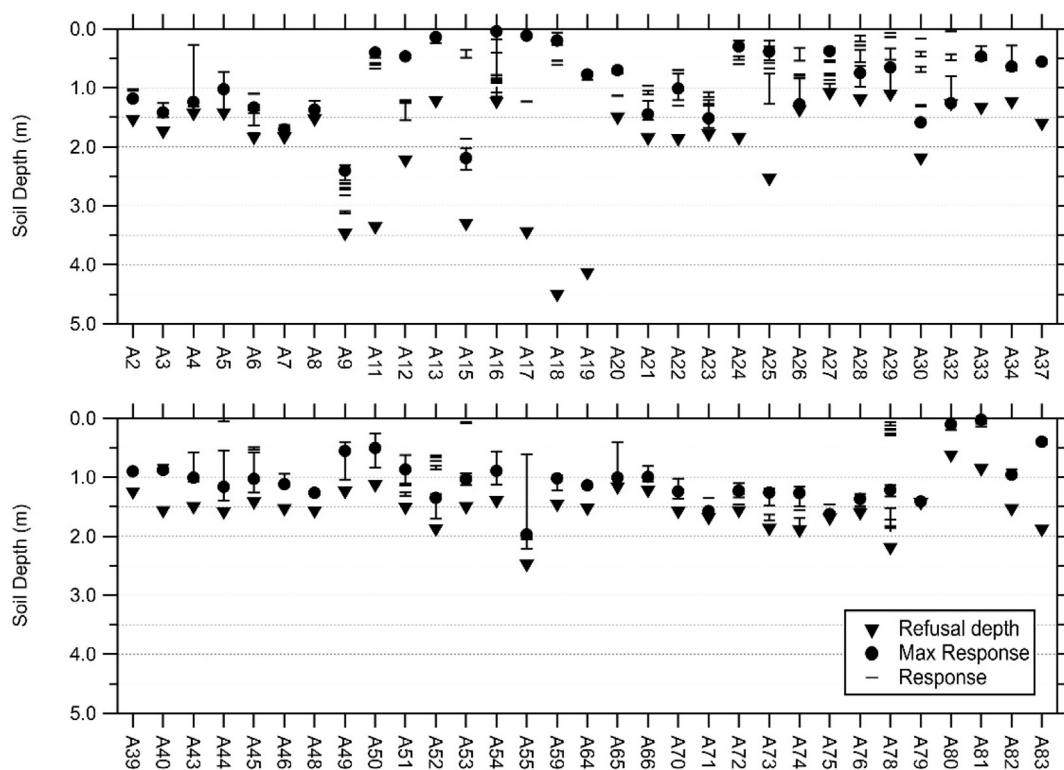


Fig. 4. Vertical PAH thickness plots for the Airstrip site. Filled circles correspond to the soil depth with the maximum fluorescence response (%) with total response represented by the solid line (above 1% Total RE). The filled triangle corresponds to the probing refusal depth at each location. Note: Probing locations with total RE % values < 1% are not shown.

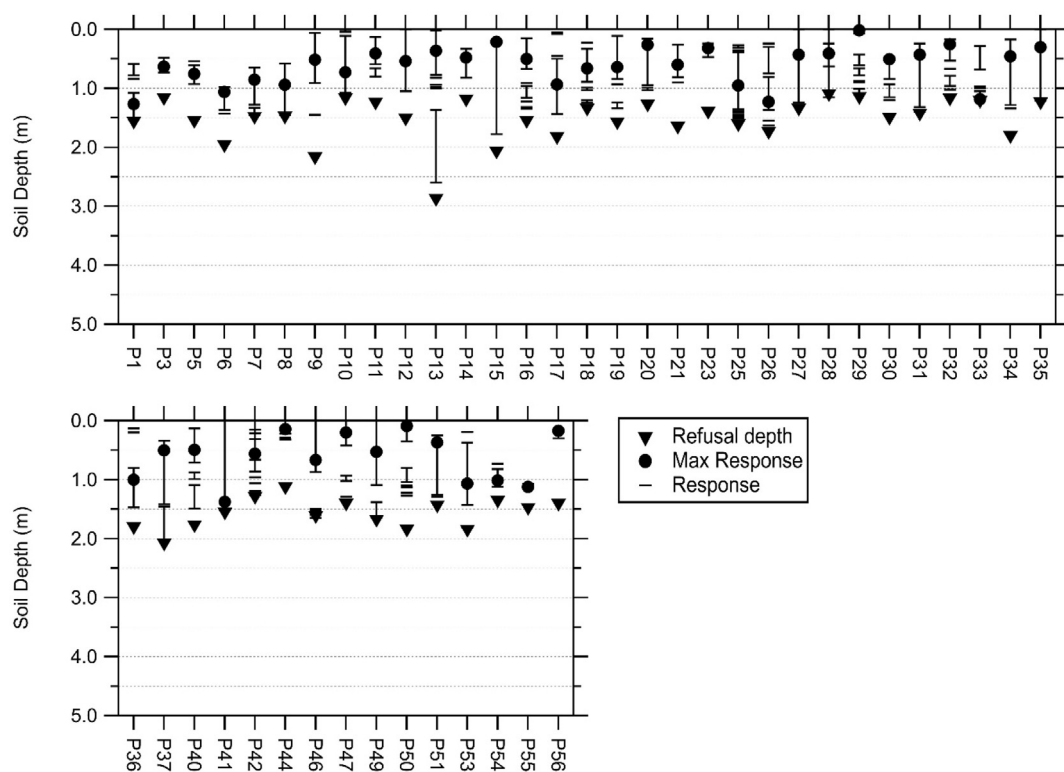


Fig. 5. Vertical PAH thickness plots for the Powerhouse site. Filled circles correspond to the soil depth with the maximum fluorescence response (%) with total response represented by the solid line (above 1% Total RE). The filled triangle corresponds to the probing refusal depth at each location. Note: Probing locations with total RE % values < 1% are not shown.

Table 1

Percentages of the location of the maximum fluorescence intensity signal (% Total RE) found within the surface to 0.5 m or deeper than 0.5 m.

Site	Surface to 0.5 m (%)	Deeper than 0.5 m (%)
Airstrip	21	79
Powerhouse	34	66

in samples that were in the same proximity of UVOST borings that also showed no measurable %RE (A18, A36, A37, A57, P25, and P52). At several locations where aromatic hydrocarbon (carbon chain length: C12–C34) was detected with the GC-FID there was measurable %RE from borings at both the Airstrip and Powerhouse sites (A43, A44, A45, A65, A66, A26, P15, P34, P35, P37, P39, P40). While the two PAH analytical quantification methods (UVOST and GC-FID) show highly comparable results between detecting the presence of PAHs or lack thereof, further work is needed to quantify the extent to which the methods are comparable across various study sites and concentration ranges of PAHs. The Powerhouse sampling locations P50 and P51 are located within a distance of 3 m and show significantly different fluorescence signals (Fig. 3). P50 exhibited a %RE max of 3.9% instead of > 12.5%, which is shown on the map. The IDW interpolation did not account for large differences in significantly close proximities and the P50 and P51 probing location was the only probing location that IDW interpolation did not represent accurately. The max fluorescence intensity at sampling location P50 is located at the surface, whereas the max fluorescence intensity at sampling location P51 is at approximately 0.5 m. Table 1 shows the percentages of sampling locations that exhibit maximum fluorescence intensities (%) from the surface to 0.5 m in comparison to depths from 0.5 m to deeper in the soil profile. Both the Airstrip and the Powerhouse sites contain a higher percentage of sampling locations that contained a fluorescence maximum deeper than 0.5 m. In particular, the Airstrip site contained 79% of the sampling locations with maximum fluorescence intensities deeper than 0.5 m. The spatial distribution and vertical thickness of the PAH plumes at the Airstrip site and Powerhouse site shows significant petroleum-derived contamination from known spills, but also shows evidence of potentially unreported spills or the potential of widespread transport of PAHs in the subsurface.

3.2. Probing refusal soil depths

The probing refusal depths were determined during the field season (2015) of this study and showed variability within each study site and when comparing across the Airstrip and Powerhouse sites. In this study, probing refusal soil depths were determined in September when the active layer is likely at the deepest yearly extent (Hinzman et al., 2005; Peterson et al., 2002). The push drill rig stopped when the probe encountered soil that was too difficult to drill through. Therefore, the terminology chosen to be used denotes ‘probing refusal depth’ instead of ‘permafrost depth’. All the PAH fluorescence measurements in this study are thought to be confined to within the active layer and do not exceed the probing refusal depths. Table 2 shows the percentages of maximum fluorescence signals that were present < 0.5 m from the

Table 2

Percentages of the location of the maximum fluorescence intensity signal (% Total RE) found < 0.5 m from the probing refusal depth or > 0.5 m from the probing refusal depth.

Site	< 0.5 m from refusal depth (%)	> 0.5 m from refusal depth (%)
Airstrip	50	50
Powerhouse	13	87

refusal depth and > 0.5 m from the refusal depth. One of the objectives of this study was to differentiate between PAH distributions that accumulated within 0.5 m above the permafrost as compared with PAH distributions that are located primarily within the active layer. The Airstrip site contained equal distributions of probing locations with the maximum fluorescence signal originating from 0.5 m above the probing refusal depth and within 0.5 m (Table 2). In contrast, the Powerhouse site contained a significantly greater amount (87%) of probing locations containing the maximum % fluorescence signal > 0.5 m from the probing refusal depth.

The presence of permafrost or any type of frozen subsurface features have been thought to act as a strong barrier for transport of elements between the surface and underlying horizons, particularly when the frozen soil is primarily composed of mineral soil layers that have a lower water storage capacity than organic soil layers (Bagard et al., 2011; Bagard et al., 2013; Keller et al., 2010; Pokrovsky et al., 2016). However, the transport of elements within frozen soil can potentially be achieved by a variety of mechanisms, including upward diffusion (wicking effect) (Noshadi et al., 2017), diffusion and thermal diffusion resulting from the existence of concentration and temperature gradients (Chuvilin, 1999), and advection, adsorption, and the downward percolation through ice-poor regions. Chuvilin (1999) showed that the transport of ions in frozen soils is primarily affected by grain size, chemical/mineral composition, moisture content, and cryogenic structure. Evidence of chemical reactivity at the permafrost-active layer boundary and the presence of an enriched, soluble layer sitting perched near the active layer-permafrost boundary have been previously observed (Barker et al., 2014). Therefore, an objective of this research was to qualitatively determine if there was any preferential accumulation of PAHs in close proximity to the probing refusal depth as a function of site. No statistical correlation was found using linear regression analysis (p -values > .05) between maximum PAH accumulation and depth of the probing refusal location at either site (Fig. 6).

3.3. Fluorescence intensity variations and implications for PAH transport

The vertical soil profiles for PAH fluorescence indicate a substantial accumulation of PAHs in the subsurface at the former NARL site at varying soil depths (Fig. 7). Individual fluorescence data are shown for 350, 400, 450, and 500 nm wavelengths of energy with the total % fluorescence intensity (response) shown as the sum of the fluorescence signal of the 4 individual wavelengths. The select locations chosen to plot at the Airstrip site represent the samples with the highest fluorescence response (with total fluorescence above 5% response). The

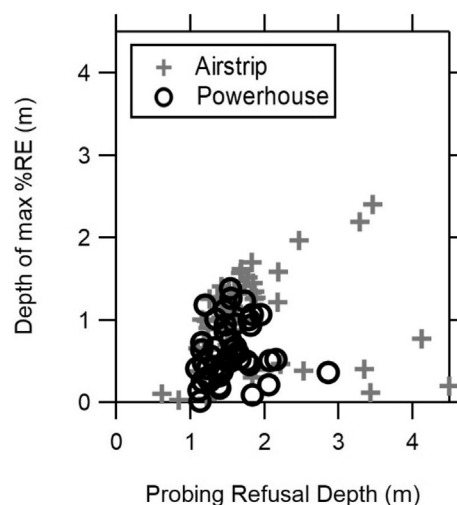


Fig. 6. Correlation plot relating the depth of the maximum fluorescence intensity signal (% Total RE) with the probing refusal depth.

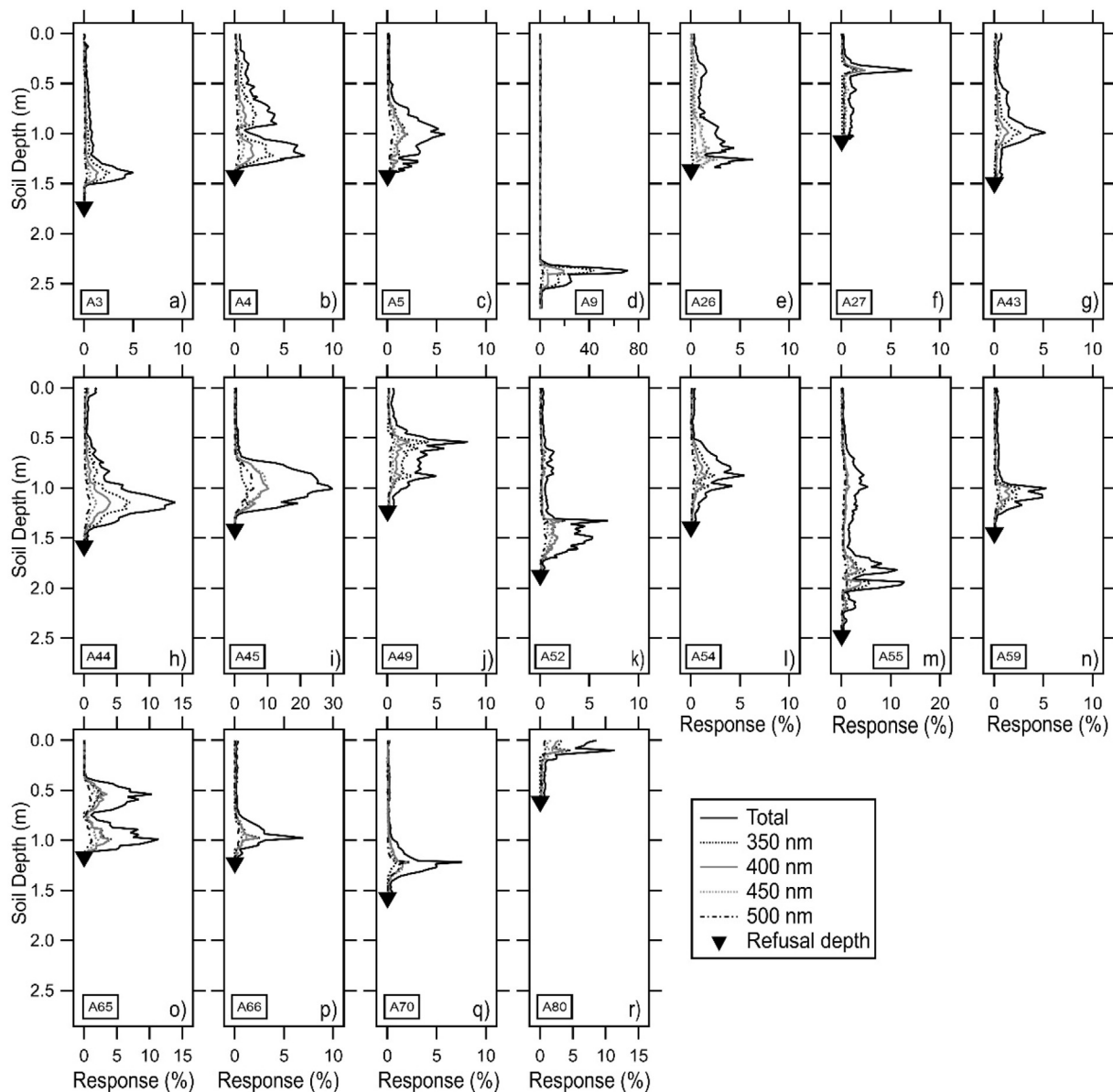


Fig. 7. Vertical profiles for PAH fluorescence signal as a function of soil depth in select sampling locations at the Airstrip site detected by LIF-UVOST in September 2015. Individual fluorescence channels are shown for 350, 400, 450, and 500 nm wavelengths of energy. Values are shown in percentages because they are normalized to a standardized fuel reference developed by Dakota Technologies. Total % fluorescence intensity (response) is the sum of the fluorescence signal of the 4 individual wavelengths. The probing refusal depth is plotted for each sampling location as a filled triangle. In vertical profile d), “*” corresponds to the probing refusal depth at A9 sampling location is at 3.5 m depth and not shown on plot.

probing locations with the greatest fluorescence signal are A9 (located on the Airstrip Runway) and A45 (located about 59 m southwest of the Airstrip Runway). Both A9 and A45 probing locations are characterized by an enhanced accumulation of PAHs at one location in the soil profile depths. Other probing locations that are similar and consist of a singular, unimodal distribution of the PAH fluorescence data are A3, A5, A27, A43, A44, A66, and A70. The singular PAH accumulations in the soil profile are found at a variety of locations within the Airstrip site. Other types of PAH distributions that were found as a function of soil depth at the Airstrip site were bimodal and multimodal distributions. Probing locations A4, A54, A65, and A80 appear to have two local maximums in the distributions, while probing locations A26, A49, A52, A55, and A59 have more than two localized enhancements of PAHs in the soil profile (Fig. 7). For probing locations A27 and A80, there is a significant enhancement of PAH distributed directly at the surface (A80) and within the top 0.4 m of soil depth.

The Powerhouse site can be characterized as containing a greater number of locations with relatively enhanced PAH accumulation (above 5% total fluorescence signal) than the Airstrip sites. While the Airstrip site contains one probing location with significantly high fluorescence intensity signal (A9), the Powerhouse site contains many more locations with PAH total fluorescence intensities above 5% (Fig. 8). The probing location with the maximum PAH fluorescence intensity was found at the Powerhouse site; P17 (Fig. 8). Probing location P17 is located between Building 442 and a road west of Imikpuk Lake (Fig. 3) and consists of a unimodal PAH enhancement at 1.0 m vertical soil depth. The probing locations with similar unimodal distributions of PAHs include P3, P5, P7, P14, P17, P18, and P32. However, a greater number of vertical PAH distributions at the Powerhouse site consisted of two or more localized distributions, including probing locations P8, P10–13, P15, P23, P26–28, P31, P33–34, P40, P41, P46, P49, and P51 (Fig. 8). Interestingly, probing locations P3, P5, and P7

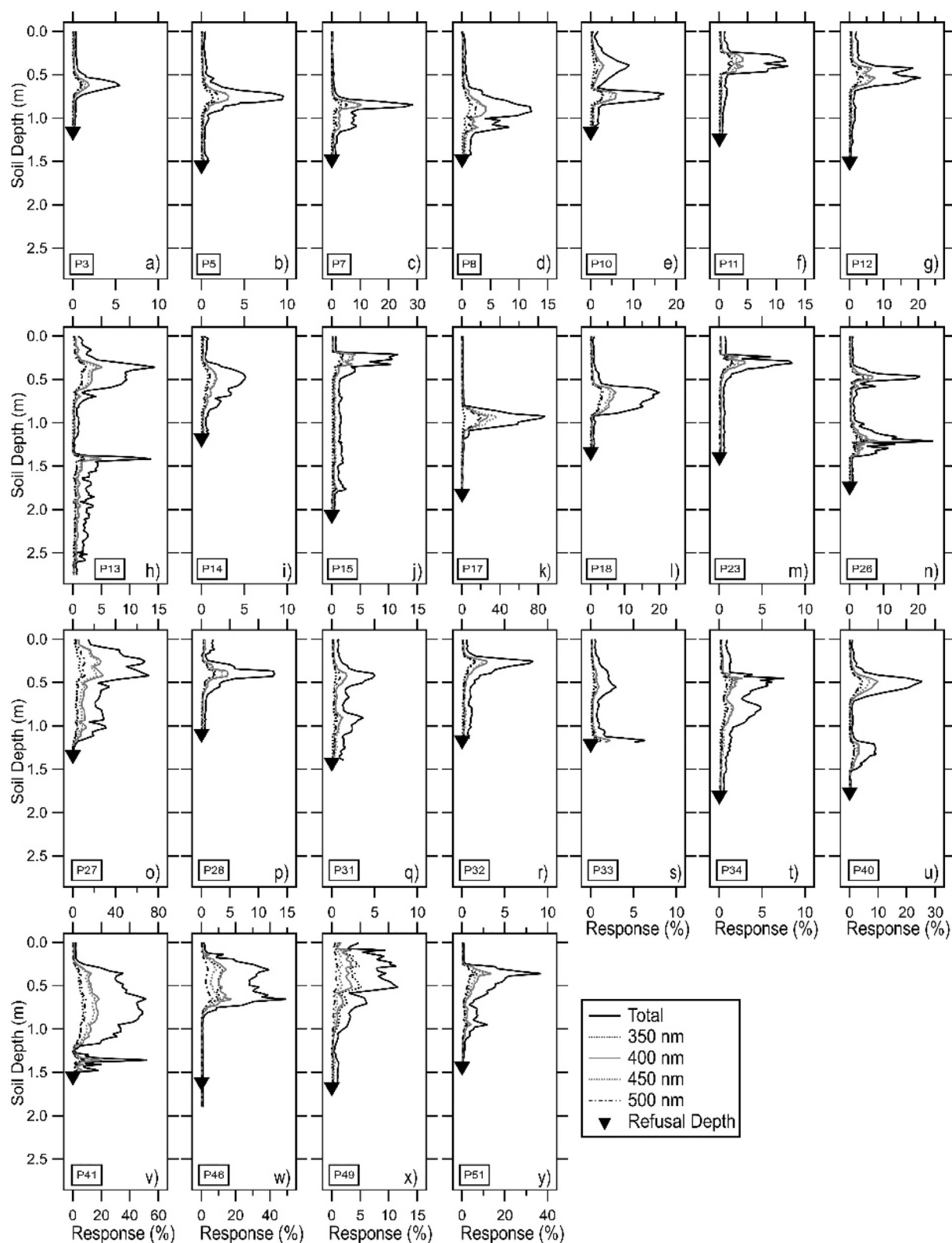


Fig. 8. Vertical profiles for PAH fluorescence signal as a function of soil depth in select sampling locations at the Powerhouse site detected by LIF-UVOST in September 2015. Individual fluorescence channels are shown for 350, 400, 450, and 500 nm wavelengths of energy. Values are shown in percentages because they are normalized to a standardized fuel reference developed by Dakota Technologies. Total % fluorescence intensity (response) is the sum of the fluorescence signal of the 4 individual wavelengths. The probing refusal depth is plotted for each sampling location as a filled triangle. In vertical profile d), “*” corresponds to the probing refusal depth at P13 sampling location is at 2.9 m depth and not shown on plot.

are all in a row located west of Imikpuk Lake arranged in order of increasing distance from the Lake (Fig. 3) and they exhibit similar PAH distributions with the only difference being depth below ground (Fig. 7). For the most part, however, the PAH fluorescence intensities were varied across the site.

The various types of PAH fluorescence distributions as a function of soil depth could be indicative of different petroleum spills over time, which is likely for probing locations A9 and A45 (Airstrip Runway) based on historical spill data. However, the differences in PAH distribution could also be a function of variations in soil type across the site, groundwater flow paths, and speciation of PAHs. The PAH fluorescence at 350, 400, 450, and 500 nm wavelengths (Figs. 7 and 8) reveal distinct patterns and signature distributions of PAHs at the Airstrip and Powerhouse sites. A study done by Wu et al. (2014) determined characteristic spectra among mixtures of standard fuel products using LIF fluorescence spectroscopy, similar to the methods used by this study. Diesel and marine fuel were found to fluoresce more intensely than jet fuel and unleaded gasoline is characterized by a principal fluorescence response at 350 nm versus 500 nm. Additionally, Wu et al. (2014) found that light-oil types of fuels produced key fluorescence responses when probed with 350 nm wavelengths of energy as opposed to 400 or 450 nm for middle-oil type fuels and 450 or 500 nm for heavy-oil type fuels. Based on this information, the results from this study found that the majority of the PAHs accumulated in the subsurface at the Airstrip site fluoresce at 350 nm to a greater extent than PAHs fluorescing at 500 nm, indicative of a light-oil type like aviation jet fuel (JP-8) or gasoline (Wu et al., 2014). When integrating the individual fluorescence channels for 350 and 500 nm (Fig. 9) only 3 locations out of the 18 probing locations at the Airstrip (with total %RE above 5%) exhibited a more pronounced fluorescence signal at 500 versus 350 nm (A26–27, and A45). Probing locations A26 and A27 are located on the banks of Imikpuk Lake, south of the Airstrip Runway and probing location A45 is located 59 m west from the Airstrip Runway. There was one probing location at the Airstrip site that contained equal contributions from the 350 and 500 nm integrated fluorescence signals (A70), which is located southeast of the Airstrip Runway. Based on the

relative distributions of the integrated fluorescence intensity values, probing location A70 could be comprised of a mixture of light and heavy-oil type fuels. The study notes that PAH degradation could also be occurring in the subsurface over time and further work is needed to quantitatively characterize the known source of spill versus degradation product at each probing location.

The integrated fluorescence signature distribution results for the Powerhouse site were the opposite as what was found at the Airstrip site, in that the majority of probing locations produced a higher response when probed by 500 nm wavelength energy (Fig. 9), indicative of a heavy-oil type fuel, like low sulfur-fuel (Wu et al., 2014). There were only seven probing locations out of 25 that contained a greater fluorescence response at 350 than 500 nm at the Powerhouse site (P23, P26, P28, P40, P46, P49, and P51). Equal contributions of both 350 and 500 nm integrated fluorescence intensity were found at 2 locations at the Powerhouse site (P3 and P34). Both the Airstrip and Powerhouse sites of the former NARL complex experienced known petroleum-based fuel spills. The detection of a higher number of probing locations fluorescing at 350 nm supports the major source of the spill was a light-oil type of fuel, like aviation fuel. Whereas, the purpose of the Powerhouse site historically was to house the Powerhouse (Building 342; Fig. 3) and the water pump house (Building 442; Fig. 3), in addition to other manufacturing buildings. The occurrence of a greater proportion of 500 nm signal is likely attributable to heavy-oil type of fuel that leaked into the subsurface at the Powerhouse. In contrast, the Airstrip likely experienced more large-scale spills from lighter-oil type fuels on and around the Airstrip Runway.

Once the fuels were released into the environment at the Airstrip and the Powerhouse sites the migration would be affected by groundwater flow paths at the site. In a previous hydrological study of the former NARL complex area, Braddock and McCarthy (1996) found three defining surface-water drainage basins: the northwestern part drains to the Arctic Ocean, the southwestern part drains to Imikpuk Lake, and the eastern part drains to North Salt Lagoon. Another report done at the former NARL complex using tracer dyes (Navy, 2018) found that groundwater flow-paths were naturally occurring to flow towards

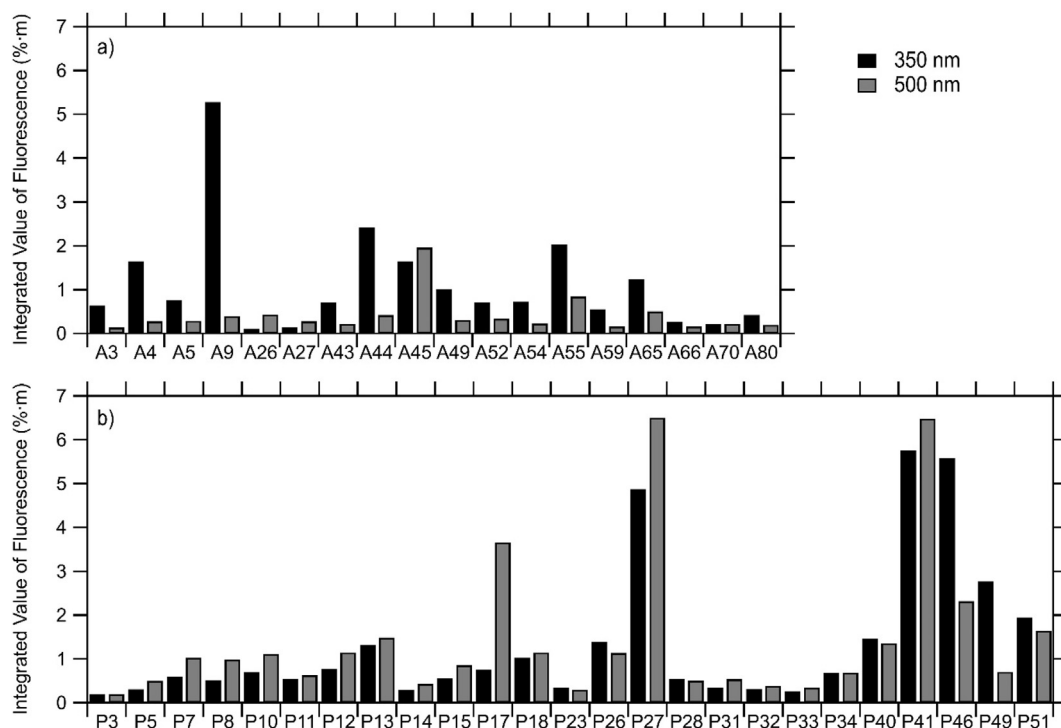


Fig. 9. Integrated values (area under the curve) of the individual fluorescence channels (350 and 500 nm) for the probing locations at the a) Airstrip and b) Powerhouse sites where total %RE was above 5%.

Imikpuk Lake at both the Airstrip and Powerhouse sites (Navy, 2018). Probing location A27 and A80 have PAH accumulated near or at the surface (Fig. 7) and are both located in close proximity to Imikpuk Lake, particularly probing location A27 (Fig. 2). While this study did not characterize PAH accumulation in Imikpuk Lake, there are significant implications for contamination and further work is needed to characterize the transport of PAHs to Imikpuk Lake as a function of subsurface distribution and site-specific soil and groundwater chemistry. Additionally, the soils at the site have been previously characterized by Barbato et al. (2017) and found to be primarily sand and gravels with some higher silt-content layers. Further work is needed to constrain site-specific soil features that may or may not be contributing to the accumulation of target PAH species in the subsurface.

4. Conclusions

The baseline results presented in this study show a substantial accumulation of PAHs in the surface and subsurface at the former NARL site near the city of Utqiagvik, Alaska. Direct chemical sensing of PAHs using LIF-UVOST was a useful tool for the in-situ determination of fluorescence intensity of PAHs in the surface and subsurface and provided information on the site-specific distribution of PAHs between two different study sites. LIF-UVOST results were compared to GC-FID results generated from a parallel study done by Barbato et al. (2017) and showed comparable results for presence of PAHs or lack thereof. Specifically, the results presented in this study and confirmed using GC-FID (Barbato et al., 2017) show that the spatial and vertical distribution of PAHs was primarily heterogeneous and varied as a function of probing locations and soil depth. There was also a distinct signature PAH fluorescence response between 350 and 500 nm intensities between the Airstrip and Powerhouse sites. The results from this study highlight LIF-UVOST analysis as an emerging, rapid response tool for the detection of PAHs in a wide range of soil types and the analytical method shows particular utility in cold environments. Future studies can use the baseline results presented here to better assist with more detailed site-wide evaluations, characterize the PAHs or other petroleum-derived hydrocarbons present, and test the efficacy of bioremediation or inorganic remediation.

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